1 Basics

This famous Einstein's mass-energy equation,$E = m.c^2$, should be rendered inline. However the following equation of motion should be rendered in a block:

$$v = u + a.t$$

2 Fraction & Roots

$$x_i = \sqrt{\frac{a^2 + b^2}{c_i}}$$

3 symbols & Math-fonts

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \qquad \forall x \in \mathbb{R}\ \exists \epsilon > 0 \Rightarrow |x| < \epsilon$$

$$F(\theta) = \sum_{k=0}^{n} \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

4 Calculus

$$\frac{du}{dx} + frac\partial u\partial t = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\int_0^1 (3x^2 + 2x + 1)\, dx$$

5 Multi-line equation

5.1 Centering & alignment

$$x = 5 \tag{1}$$

$$x + y = 3 \tag{2}$$

$$x = 5 \tag{3}$$

$$x + y = 3 \tag{4}$$

5.2 Multiline equation

$$(x+y+z)^3 = (x+y+z)(x+y+z)(x+y+z) \quad = x^3 + y^3 + z^3 + 3x^2y + 3x^2z + 3y^2z + 3zy^2 + 3xy^2 + 3zx^2 + 6xyz$$

6 Matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \quad \begin{Bmatrix} 1 & 2 \\ 3 & 4 \end{Bmatrix}$$